

Constantinos Xenophontos

Curriculum Vitae



Postdoctoral researcher, IRCCS Humanitas Research Hospital, Milan
c_xen@outlook.com · orcid.org/0000-0001-9340-9566
github.com/cxen · cxen.github.io

Employment

Since 2023 **Postdoctoral researcher**, IRCCS Humanitas Research Hospital, Milan, Italy
PI: Dr. Gabriele Micali, Microbial Ecology Lab

Education

2016 – 2020 **PhD researcher**, Friedrich Schiller University Jena & German Centre for Integrative Biodiversity Research (iDiv) Halle-Jena-Leipzig, Germany
Dissertation: *Testing ecological theories with microbial communities*.
Supervisors: Prof. Dr. Kirsten Küsel, Prof. Dr. W. Stanley Harpole.

2014 – 2015 **MRes Ecology, Evolution & Conservation (Merit)**, Imperial College London, UK
Thesis 1: *Variation in behaviour and resistance to fungal endoparasites between closely related clones of the anciently asexual rotifer Adineta vaga*.
Thesis 2: *Effects and interactions on phenotypic trajectories and metabolic activity of community bacteria*.

2010 – 2014 **BSc (Hons) Biology (2:1)**, University of Hull, UK
Team research project in ecology, DNA barcoding and bioinformatics: *Is soil mesofauna abundance and diversity related to farming methods?*

Publications

Xenophontos, C., Harpole, W. S., Küsel, K., & Clark, A. T. (2022). Cheating promotes coexistence in a two-species one-substrate culture model. *Frontiers in Ecology and Evolution*, 9, 786006. <https://doi.org/10.3389/fevo.2021.786006>

Xenophontos, C., Taubert, M., Harpole, W. S., & Küsel, K. (2021). Phylogenetic and metabolic diversity have contrasting effects on the ecological functioning of bacterial communities. *FEMS Microbiology Ecology*, 97(3), fiab017. <https://doi.org/10.1093/femsec/fiab017>

The 2021 *FEMS Microbiology Ecology* article was selected as an Editor's Choice.

Conference presentations

2026 May **Microfluidic Horizons 2026**, oral presentation
Stronger at the seams: antibiotic stress enhances amino acid leakage and cross-feeding.
Xenophontos C., Vaidyanathan A., Micali G.

2025 Dec	BES Annual Meeting 2025 , poster presentation <i>Stronger at the seams: antibiotic stress enhances amino acid leakage and cross-feeding.</i> Xenophontos C., Vaidyanathan A., Micali G.
2025 Jul	FEMS Micro 2025 , oral and poster presentations <i>Stronger at the seams: antibiotic stress enhances amino acid leakage and cross-feeding.</i> Xenophontos C., Vaidyanathan A., Micali G.
2020 Dec	ISME Virtual Microbial Ecology Summit , poster presentation <i>Coexisting with cheaters: microbial cheating as a snowdrift game model.</i> Xenophontos C., Harpole W.S., Küsel K., Clark A.T. doi:10.13140/RG.2.2.36728.26886
2019 Dec	BES Annual Meeting 2019 , oral presentation <i>Phylogenetic and functional dissimilarity have contrasting effects on the ecological functioning of synthetic bacterial communities.</i> Xenophontos C., Taubert M., Harpole W.S., Küsel K. doi:10.13140/RG.2.2.16123.67361
2019 Sep	SAME16 , poster presentation <i>Species phylogenetic and functional diversity contribute to ecological functioning of groundwater bacterial communities.</i> Xenophontos C., Taubert M., Harpole W.S., Küsel K.
2019 Aug	5th iDiv Conference , poster presentation <i>Species phylogenetic and functional diversity contribute to ecological functioning of groundwater bacterial communities.</i> Xenophontos C., Taubert M., Harpole W.S., Küsel K.
2018 Dec	4th iDiv Conference , oral presentation <i>Species phylogenetic and functional diversity contribute to ecological functioning of groundwater bacterial communities.</i> Xenophontos C., Taubert M., Harpole W.S., Küsel K.
2018 Aug	ISME17 , poster presentation <i>Species phylogenetic and functional diversity contribute to ecological functioning of groundwater bacterial communities.</i> Xenophontos C., Taubert M., Harpole W.S., Küsel K.
2017 Sep	3rd iDiv Conference , poster presentation <i>Shifts in ecological functioning of groundwater bacteria driven by species interactions.</i> Xenophontos C., Geesink P., Kumar S., Wegner C.-E., Harpole W.S., Küsel K. doi:10.13140/RG.2.2.24728.49929

Teaching and supervision

Since 2023	Co-supervision of PhD and MSc students with Dr. Micali (lab work and data analysis), IRCCS Humanitas Research Hospital
2019 May	Soil DNA extraction and PCR practical course, Friedrich Schiller University Jena

Academic service

2026	Microfluidic Horizons 2026 , organising committee Social media and communications; session moderator
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Outreach

2023	<i>Expo per lo Sport</i> , public event, Humanitas booth, Milan DNA-extraction instructor for children
2022	<i>Holobiontinnen</i> , art performance exhibit, Leipzig (LOFFT, Ballhaus Ost) Consulting microbiologist
2019	<i>Long Night of Sciences</i> , public event, Jena Instructor, microbiology art for children

Grants, awards and studentships

2023	EMBO Postdoctoral Fellowship, shortlisted and interviewed; not funded
2023	HFSP Long-Term Fellowship, applied; not funded
2019	SAME16 conference travel grant
2016	yDiv Doctoral Programme studentship, German Centre for Integrative Biodiversity Research (iDiv) Halle-Jena-Leipzig

Societies

- International Society for Microbial Ecology (ISME), member
- Royal Society of Biology (RSB), member (MRSB)
- British Ecological Society (BES), member
- Federation of European Microbiological Societies (FEMS), affiliate member
- Society for Open, Reliable, and Transparent Ecology and Evolutionary biology (SORTEE), member

Skills

- **Experimental:** bacterial growth assays in 96-well plate readers, microfluidic timelapse microscopy, continuous-culture bioreactors, synthetic bacterial communities, DNA extraction and PCR
- **Computational:** R (tidyverse, Quarto), Python (marimo, Polars), Julia; growth-curve and dose-response analysis; image segmentation and cell tracking (Omnipose); Git and GitHub
- **Hardware and fabrication:** open-source bioreactor firmware (ChiBio); 3D printing and design (PrusaSlicer, BambuStudio, Fusion 360, Meshmixer)